

# Rahul Rawat

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**1KOMMA5°**

*Hiring Team*

## **Application: Working Student, Data Science Forecasting**

Dear Hiring Team,

I am applying for the Working Student position in Data Science Forecasting at 1KOMMA5°. The role reads like the natural next step from the recursive forecasting pipeline I built at eRay GmbH with SRH University Heidelberg, where I forecasted chlorophyll a, turbidity, pH, and dissolved oxygen for a German lake over six months. I trained and compared six models head to head, including CatBoost with multi quantile regression for asymmetric 80 percent intervals, and wrapped the pipeline in an orchestrator with gate checks that halted on failed imputation instead of letting bad data cascade into the forecast.

The mix of PV, load, heat pump, and EV charging signals in your Heartbeat AI stack is exactly the kind of asymmetric, physically constrained forecast problem I have been training for. In my flight tracking project I built a streaming pipeline that polled live positions every 30 seconds, cleaned data with PySpark on Google Cloud, and shaped analytics ready tables with dbt in BigQuery. The Movie Analytics medallion pipeline I built added BigQuery ML with strict feature separation to avoid data leakage. Both projects map directly onto SQLMesh style transformations, backtesting, and production monitoring.

I have also built a modular Retrieval Augmented Generation system on top of LangChain, ChromaDB, and Llama 3.1 8b through Groq, with a stateful memory agent that maintains conversational context. That work translates well to the agentic AI workflow automation angle in your role description.

I would be glad to discuss how I could support your forecasting team, starting from either the model training side or the production pipeline side, whichever helps you fastest. Thank you for considering my application.

Kind regards,

**Rahul Rawat**